

Mapping Semantic Spaces with Simple NLP Embeddings

Natural Language Processing Final Project - P15: Cartographers of the Invisible

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Abstract

This project studies whether a simple NLP representation can organize words into meaningful semantic groups. A small dataset of 50 concepts was created across five categories: sport, food, profession, emotion, and technology. Each concept is represented using a short textual description. The text is converted into TF-IDF embeddings, clustered using K-Means, and visualized using Principal Component Analysis (PCA). The goal is not to reach state-of-the-art performance, but to build a clear and reproducible experiment that shows how numerical text representations can capture semantic similarity. The results show partial semantic organization: some categories form visible groups, while others overlap due to ambiguity and the limits of the simple representation.

1. Introduction

Natural Language Processing often requires converting language into numerical vectors that can be processed by algorithms. These representations are useful because they allow words, sentences, and documents to be compared mathematically. In modern NLP, embeddings are frequently used to study semantic similarity, search, clustering, and visualization of language data.

The selected project topic is P15, Cartographers of the Invisible. The general idea is to make semantic spaces visible by extracting embeddings and projecting them into two-dimensional maps. This simplified version focuses on individual concepts rather than large language models. The experiment uses a compact dataset and classical NLP methods so that the full workflow remains easy to reproduce and explain.

The research question is: Can simple text embeddings automatically organize concepts into meaningful semantic groups?

2. Research Question and Methodology

The objective is to test whether concepts that belong to the same human-defined category are placed close to each other in a vector space. The measurable goals are: (1) cluster concepts based only on their text representation, (2) compare predicted clusters with the original categories, and (3) visualize the semantic space using PCA.

Dataset. The dataset contains 50 concepts divided equally into five categories. Each row contains a concept, a category label, and a short description. The categories were selected because they are easy to understand and semantically different: sport, food, profession, emotion, and technology.

Example concepts include football, basketball, tennis and cricket for sport; pizza, pasta and coffee for food; doctor, teacher and programmer for professions; happiness, sadness and anger for emotions; and computer, internet and database for technology.

Text representation. The concept and its description are combined into a single document. TF-IDF is used to transform each document into a numerical vector. TF-IDF gives more importance to informative terms and less importance to very common terms. Although it is simpler than transformer embeddings, it is enough for a baseline semantic mapping experiment.

Clustering. K-Means is applied to the vectors with $k = 5$, matching the number of known categories. The algorithm assigns concepts to clusters based on vector similarity. The clustering is then compared with the original category labels.

Dimensionality reduction. PCA reduces the high-dimensional TF-IDF vectors into two dimensions. This makes it possible to plot the concepts on a two-dimensional map. PCA is used only for visualization, not for the main clustering evaluation.

Evaluation metrics. Three metrics are used: Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Silhouette Score. ARI and NMI compare predicted clusters with the original labels. Silhouette Score measures how separated the clusters are in the vector space.

3. Experimental Results

The experiment was executed using the Python script `src/main.py`. The script loads the dataset, creates TF-IDF embeddings, applies K-Means clustering, computes evaluation metrics, and saves the PCA plot and result files.

Metric	Value
Adjusted Rand Index	0.5222
Normalized Mutual Information	0.7073
Silhouette Score	0.0250

Table 2. Clustering evaluation results.

The ARI and NMI scores indicate that the model recovered part of the original category structure. The NMI score is higher because the clustering captures some information about the true categories even when the exact cluster assignments are not perfect. The Silhouette Score is low, which means that the clusters are not strongly separated in the TF-IDF vector space.

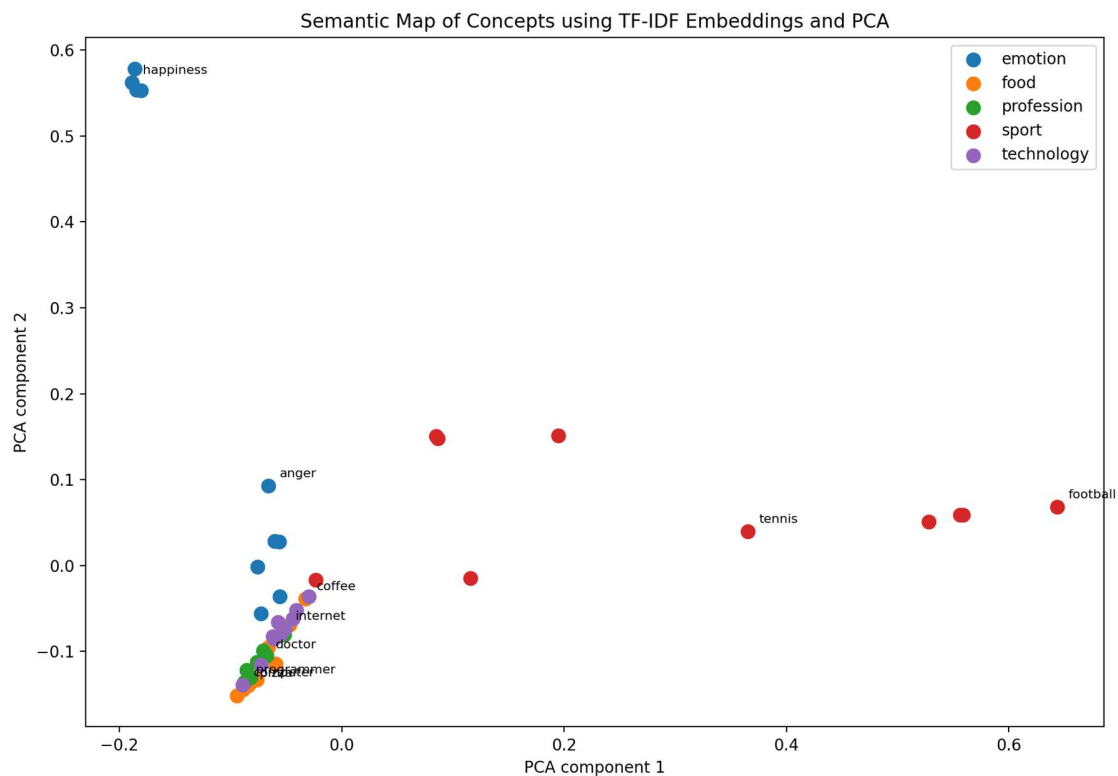


Figure 1. PCA visualization of the semantic space.

The visualization shows that some groups, especially sport and emotion concepts, form recognizable areas. However, several food, profession, and technology concepts overlap. This is an important result because it shows both the usefulness and the limitations of a simple embedding method. Concepts can be related semantically, but short descriptions and TF-IDF vectors do not always capture deep meaning.

4. Concluding Remarks

The project confirms that simple NLP embeddings can partially organize concepts into meaningful semantic groups. The method is easy to reproduce and provides clear outputs: clusters, evaluation metrics, and a visual semantic map. The results are not perfect, but this is expected because TF-IDF is based mainly on word overlap and does not understand meaning in the same way as modern transformer models.

The main limitation is the small manually created dataset. Another limitation is that the descriptions influence the results strongly: if two concepts have similar descriptions, they may appear closer even if they belong to different categories. PCA also loses information when reducing high-dimensional vectors to two dimensions.

Future work could replace TF-IDF with Sentence-BERT or another transformer model, use a larger dataset, compare multiple dimensionality reduction methods such as t-SNE and UMAP, and add an interactive visualization with Streamlit or Gradio.

AI Usage Disclaimer

Parts of this project were developed with assistance from OpenAI's ChatGPT. The AI was used to support the organization of the project structure, drafting of explanatory text, and preparation of simple Python code. All

content produced with AI assistance has been carefully reviewed, edited, and validated by me. I take full responsibility for the final content and its accuracy, relevance, and academic integrity

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